

# The Complete Asymptotic Transseries of the Fubini Numbers

## Exact Pole Sectors, All-Orders Coefficients, Certified Remainders, and a Resurgent Interpretation

A self-contained analytic-combinatorial synthesis

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### Abstract

The Fubini numbers, also called ordered Bell numbers or preferential-arrangement numbers, have exponential generating function  $(2 - e^z)^{-1}$ . Its singularities form the vertical lattice  $\rho_k = \log 2 + 2\pi i k$ ,  $k \in \mathbb{Z}$ , and every singularity is a simple pole of residue  $-1/2$ . This elementary meromorphic geometry makes it possible to go much further than the familiar leading estimate  $F_n \sim n!/[2(\log 2)^{n+1}]$ . For every  $n \geq 1$  we prove the exact, absolutely convergent identity

$$F_n = \frac{n!}{2} \sum_{k \in \mathbb{Z}} (\log 2 + 2\pi i k)^{-n-1}.$$

After normalization by the dominant pole, this is already the complete exponential–oscillatory transseries: every nonzero pole gives one sector, no sector carries an inverse-power descendant tower, and there are no hidden Fubini-specific sectors. Pairing conjugate poles gives a real expansion whose amplitudes, actions, and phases are explicit.

We derive explicit and elementary truncation bounds in terms of the Hurwitz zeta function, obtain certified nearest-integer reconstruction criteria, and explain why exponentially small relative errors may nevertheless be enormous in absolute value. Expanding the exact factorial prefactor by Stirling’s series produces the conventional two-level transseries. Its coefficient in pole sector  $k$  and algebraic order  $j$  is

$$C_{k,j} = \frac{\log 2}{\log 2 + 2\pi i k} \gamma_j,$$

where  $\gamma_j$  is given explicitly by complete exponential Bell polynomials, by a partition sum, and by a Bernoulli-number recurrence. A multi-index coefficient calculus is then developed for arbitrary powers, logarithms, reciprocals, and other analytic functions of the normalized sequence. Finally, the formal ordinary generating series is shown to have Borel transform exactly  $(2 - e^\xi)^{-1}$ , so the pole lattice is at once the Borel singularity set and the complete large-order data. Extensions to Fubini polynomials and to higher powers  $(2 - e^z)^{-r}$  are included, the latter having general coefficients expressed by Nörlund (generalized Bernoulli) numbers.

**Master result**

Put

$$\alpha = \log 2, \quad \rho_k = \alpha + 2\pi i k, \quad \omega_k = \frac{\alpha}{\rho_k}.$$

Then, for every integer  $n \geq 1$ ,

$$\frac{2\alpha^{n+1}}{n!} F_n = \sum_{k \in \mathbb{Z}} \omega_k^{n+1}$$

is an *exact* identity, not merely an asymptotic expansion. Since  $\omega_0 = 1$  and  $|\omega_k| < 1$  for  $k \neq 0$ , it is simultaneously the complete factorial-normalized transseries.

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# 1 Introduction and scope

## 1.1 From ordered partitions to a meromorphic generating function

A partition of a labelled  $n$ -element set becomes an *ordered set partition* when its blocks are arranged in a linear order. If there are  $j$  blocks, then the underlying partition can be chosen in  $\left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\}$  ways and the blocks can be ordered in  $j!$  ways. Consequently the  $n$ th Fubini number is

$$F_n = \sum_{j=0}^n j! \left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\}, \quad n \geq 0. \quad (1)$$

The first values are

$$1, 1, 3, 13, 75, 541, 4683, 47293, 545835, 7087261, \dots$$

The same sequence counts weak orderings, preferential arrangements, and a variety of equivalent structures; see Comtet [3], Bailey [1], Pippenger [8], and the references in OEIS A000670 [7].

The labelled sequence construction immediately gives

$$G(z) := \sum_{n \geq 0} F_n \frac{z^n}{n!} = \sum_{j \geq 0} (e^z - 1)^j = \frac{1}{2 - e^z}. \quad (2)$$

The linked *Combinatorial Coefficient Calculus* develops precisely this Stirling-number and exponential-generating-function foundation [9]. The linked series-and-transseries collection develops the formal operations used after an asymptotic scale has been identified [10]. The present article joins those two viewpoints: it extracts the complete transseries from the meromorphic geometry of (2), and then expresses every coefficient in the Bell-polynomial language of coefficient calculus.

## 1.2 Why an ordinary Poincaré series is not enough

The nearest singularity of  $G$  to the origin is the positive real pole  $\alpha = \log 2$ , so elementary singularity analysis [6, Chap. IV] yields

$$F_n \sim \frac{n!}{2\alpha^{n+1}}. \quad (3)$$

This classical equivalent goes back at least to Barthélemy's analysis of total preorders [2]. It suppresses, however, an infinite lattice of further poles at  $\alpha + 2\pi i k$ . Their contributions are smaller than every power of  $1/n$  relative to the dominant term, and therefore an ordinary Poincaré series sees only

$$\frac{2\alpha^{n+1}F_n}{n!} \sim 1 + \frac{0}{n} + \frac{0}{n^2} + \dots$$

The zero coefficients are correct but incomplete: the actual error is a structured sum of exponentially small, oscillatory terms. A transseries keeps those terms rather than discarding them.

## 1.3 What “full” means in this article

There are two natural normalizations, and it is useful to distinguish them.

1. **Factorial-normalized expansion.** Keep  $n!$  exact. The pole sum in the abstract is then exact and contains every Fubini-specific exponential sector. Each simple pole contributes one transmonomial and no inverse-power descendants.
2. **Elementary Stirling normalization.** Expand  $n!$  in powers of  $1/n$ . Every pole sector then inherits the same universal Stirling tower. We derive its coefficients to arbitrary order and give a closed formula for the coefficient attached to every pair  $(k, j)$  of pole index and algebraic order.

The first form is analytically stronger because it is exact. The second is the form usually expected when a transseries is written over the elementary scale  $\sqrt{n}(n/(e\alpha))^n$ . Together they leave no unspecified Fubini-specific remainder.

## 2 Combinatorial and exact analytic foundations

### 2.1 The exponential generating function and recurrence

**Theorem 2.1** (Basic Fubini identities). *For  $n \geq 0$ , the numbers in (1) satisfy*

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (4)$$

$$F_0 = 1, \quad F_n = \sum_{j=1}^n \binom{n}{j} F_{n-j} \quad (n \geq 1), \quad (5)$$

$$F_n = \frac{1}{2} \sum_{m=0}^{\infty} \frac{m^n}{2^m}. \quad (6)$$

In (6), the convention  $0^0 = 1$  is used when  $n = 0$ . For  $n \geq 1$ , the term  $m = 0$  vanishes.

*Proof.* The combinatorial derivation of (4) was given in (2). Multiplying that identity by  $2 - e^z$  and comparing the coefficient of  $z^n/n!$  gives

$$2F_n - \sum_{j=0}^n \binom{n}{j} F_{n-j} = 0.$$

The  $j = 0$  summand is  $F_n$ , and moving it to the other side gives (5). Finally,

$$\frac{1}{2 - e^z} = \frac{1}{2} \frac{1}{1 - e^z/2} = \frac{1}{2} \sum_{m \geq 0} \frac{e^{mz}}{2^m}.$$

Extracting the coefficient of  $z^n/n!$  proves (6). The interchange is valid analytically near  $z = 0$ , and also as an identity of formal exponential series because every coefficient is an absolutely convergent geometric moment.  $\square$

*Remark 2.2* (Probabilistic interpretation). Let  $X$  have the geometric distribution  $\Pr(X = m) = 2^{-m-1}$  for  $m = 0, 1, 2, \dots$ . Then  $F_n = \mathbb{E}[X^n]$ . Thus the same moments that count ordered partitions also measure powers of the number of failures before the first success in fair Bernoulli trials.

*Remark 2.3* (Polylogarithm form). For  $n \geq 1$ ,

$$F_n = \frac{1}{2} \text{Li}_{-n}\left(\frac{1}{2}\right). \quad (7)$$

This is exact, but the pole representation developed below is better adapted to large  $n$ .

### 2.2 The continuum term already gives the dominant asymptotic

Set

$$\alpha := \log 2. \quad (8)$$

The moment formula may be written

$$F_n = \frac{1}{2} \sum_{m \geq 1} m^n e^{-\alpha m}, \quad n \geq 1. \quad (9)$$

Replacing the lattice sum by its continuum integral gives

$$\frac{1}{2} \int_0^\infty x^n e^{-\alpha x} dx = \frac{\Gamma(n+1)}{2\alpha^{n+1}} = \frac{n!}{2\alpha^{n+1}}. \quad (10)$$

The remarkable fact is that the difference between (9) and (10) is not an inverse-power Euler–Maclaurin series. For fixed positive  $\alpha$ , all endpoint derivatives that could contribute at algebraic orders are encoded by an exact periodic pole sum, and the relative difference is exponentially small in  $n$ . Poisson summation will make this statement exact.

### 3 The singularity lattice and the exact pole formula

#### 3.1 Location and residues of all singularities

The poles of  $G(z) = (2 - e^z)^{-1}$  solve  $e^z = 2$ . Hence

$$\rho_k := \alpha + 2\pi i k, \quad k \in \mathbb{Z}. \quad (11)$$

They are all simple, and

$$\operatorname{Res}_{z=\rho_k} G(z) = \frac{1}{-e^{\rho_k}} = -\frac{1}{2}. \quad (12)$$

Thus every pole has the same residue; only its distance and phase from the origin vary.

#### 3.2 A Mittag–Leffler identity

The following classical partial fraction identity is the analytic engine behind the exact expansion; see also [12].

**Lemma 3.1** (Partial fractions for the Bose–Einstein kernel). *For  $w \notin 2\pi i\mathbb{Z}$ ,*

$$\frac{1}{e^w - 1} = \frac{1}{w} - \frac{1}{2} + 2w \sum_{k=1}^{\infty} \frac{1}{w^2 + 4\pi^2 k^2}. \quad (13)$$

*The series converges locally uniformly away from the poles. Equivalently,*

$$\frac{1}{2} \coth \frac{w}{2} = \frac{1}{w} + 2w \sum_{k=1}^{\infty} \frac{1}{w^2 + 4\pi^2 k^2}. \quad (14)$$

*Proof.* The canonical product

$$\frac{\sinh(w/2)}{w/2} = \prod_{k=1}^{\infty} \left( 1 + \frac{w^2}{4\pi^2 k^2} \right)$$

converges locally uniformly. Taking its logarithmic derivative gives (14). Since  $(e^w - 1)^{-1} = \frac{1}{2} \coth(w/2) - \frac{1}{2}$ , equation (13) follows.  $\square$

#### 3.3 Exact pole expansion

**Theorem 3.2** (Exact lattice formula). *For every integer  $n \geq 1$ ,*

$$\boxed{F_n = \frac{n!}{2} \sum_{k \in \mathbb{Z}} \rho_k^{-n-1}} \quad (\rho_k = \log 2 + 2\pi i k). \quad (15)$$

*The series is absolutely convergent.*

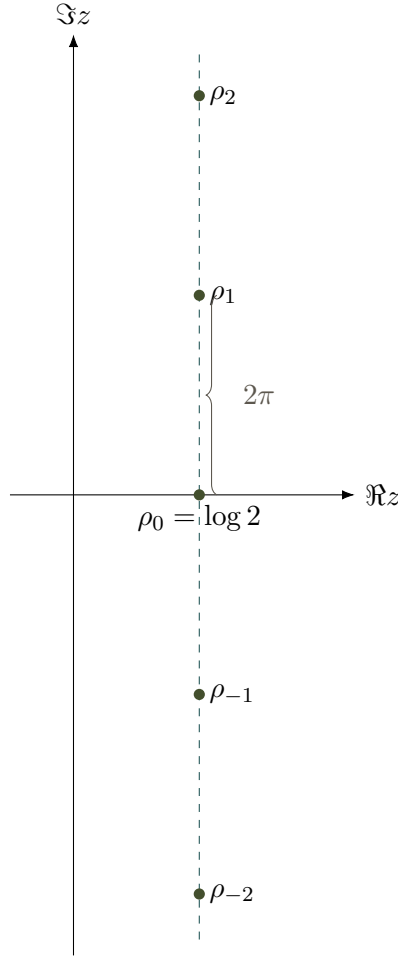


Figure 1: The complete singularity lattice of  $G(z) = (2 - e^z)^{-1}$ . Every point on the line  $\Re z = \log 2$  is a simple pole of residue  $-1/2$ .

*Proof.* From (9),

$$2F_n = \sum_{m \geq 1} m^n e^{-\alpha m} = (-1)^n \frac{d^n}{d\alpha^n} \frac{1}{e^\alpha - 1}. \quad (16)$$

Differentiate (13)  $n$  times. For  $n \geq 1$ , the constant  $-1/2$  disappears, and the differentiated partial fractions can be paired as

$$\frac{1}{\alpha + 2\pi i k} \quad \text{and} \quad \frac{1}{\alpha - 2\pi i k}.$$

Local uniform convergence justifies termwise differentiation. Since

$$(-1)^n \frac{d^n}{d\alpha^n} \frac{1}{\alpha + 2\pi i k} = \frac{n!}{(\alpha + 2\pi i k)^{n+1}},$$

substitution into (16) yields (15). Absolute convergence follows from  $|\rho_k|^{-n-1} = \mathcal{O}(|k|^{-n-1})$ , whose sum converges for  $n \geq 1$ .  $\square$

*Remark 3.3* (The exceptional index  $n = 0$ ). Formula (15) is stated for  $n \geq 1$  because the corresponding two-sided pole series is not absolutely convergent and requires symmetric summation; moreover, the constant  $-1/2$  in (13) no longer disappears. In fact, symmetric summation gives

$$\sum_{k \in \mathbb{Z}}^{\text{sym}} \frac{1}{\alpha + 2\pi i k} = \frac{1}{2} \coth\left(\frac{\alpha}{2}\right) = \frac{3}{2}, \quad \alpha = \log 2,$$

so the naively continued right-hand side of (15) is  $3/4$ , not  $F_0 = 1$ . The bookkeeping is transparent in (13): multiplying the regularization constant  $-1/2$  by the outer factor  $1/2$  changes  $3/4$  to the positive- $m$  moment  $1/2$ , and the separate  $m = 0$  term in (6) contributes the remaining  $1/2$ . Thus the exact pole identity is naturally an  $n \geq 1$  statement.

### 3.4 A second proof by one-sided Poisson summation

The pole formula also has a transparent harmonic-analysis interpretation.

**Proposition 3.4** (Poisson form of the lattice correction). *For  $n \geq 1$ , define*

$$f_n(x) = \begin{cases} x^n e^{-\alpha x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

With Fourier transform  $\widehat{f}_n(k) = \int_{\mathbb{R}} f_n(x) e^{-2\pi i k x} dx$ , one has

$$\widehat{f}_n(k) = \frac{n!}{(\alpha + 2\pi i k)^{n+1}}. \quad (17)$$

Poisson summation gives

$$\sum_{m \geq 1} m^n e^{-\alpha m} = \sum_{k \in \mathbb{Z}} \frac{n!}{(\alpha + 2\pi i k)^{n+1}}, \quad (18)$$

and hence (15).

*Proof.* The Laplace integral gives (17). The function  $f_n$  is continuous at the origin for  $n \geq 1$ , piecewise smooth, exponentially decaying, and its transform is absolutely summable because it is  $\mathcal{O}(|k|^{-n-1})$ . The standard one-sided form of Poisson summation is therefore applicable. Since  $f_n(m) = 0$  for negative integers and  $f_n(0) = 0$ , its left side is the sum in (18). Combining this with (9) proves the result.  $\square$

#### Structural point

The  $k = 0$  Fourier mode in (18) is exactly the continuum integral (10). Every  $k \neq 0$  mode is an exponentially small lattice correction. Thus the full transseries can be read either from the poles of the exponential generating function or from the nonzero Fourier harmonics of the geometric moment sum.

## 4 The complete factorial-normalized transseries

### 4.1 Actions, amplitudes, and phases

Define, for  $k \in \mathbb{Z}$ ,

$$\omega_k := \frac{\alpha}{\rho_k}, \quad \Lambda_k := \text{Log} \frac{\rho_k}{\alpha} = -\text{Log} \omega_k, \quad (19)$$

where  $\text{Log}$  is the principal logarithm. For  $k \geq 1$ , also put

$$r_k := |\rho_k| = \sqrt{\alpha^2 + 4\pi^2 k^2}, \quad (20)$$

$$q_k := |\omega_k| = \frac{\alpha}{r_k}, \quad (21)$$

$$\lambda_k := \Re \Lambda_k = \log \frac{r_k}{\alpha} = \frac{1}{2} \log \left( 1 + \frac{4\pi^2 k^2}{\alpha^2} \right), \quad (22)$$

$$\theta_k := \arg \rho_k = \arctan \frac{2\pi k}{\alpha}. \quad (23)$$

Then  $q_k = e^{-\lambda_k}$ ,  $\omega_k = q_k e^{-i\theta_k}$ , and  $\omega_{-k} = \overline{\omega_k}$ .



**Theorem 4.1** (Complete exact transseries). *For every  $n \geq 1$ ,*

$$\frac{2\alpha^{n+1}}{n!} F_n = \sum_{k \in \mathbb{Z}} \omega_k^{n+1} \quad (24)$$

$$= 1 + 2 \sum_{k=1}^{\infty} q_k^{n+1} \cos((n+1)\theta_k) \quad (25)$$

$$= 1 + 2 \sum_{k=1}^{\infty} e^{-(n+1)\lambda_k} \cos((n+1)\theta_k). \quad (26)$$

*All three expressions are exact and absolutely convergent. For every fixed  $K \geq 0$ ,*

$$\frac{2\alpha^{n+1}}{n!} F_n = 1 + 2 \sum_{k=1}^K q_k^{n+1} \cos((n+1)\theta_k) + \mathcal{O}_K(q_{K+1}^{n+1}) \quad (27)$$

*as  $n \rightarrow \infty$ .*

*Proof.* Equation (24) is just (15) multiplied by  $2\alpha^{n+1}/n!$ . Pairing the terms indexed by  $k$  and  $-k$  gives (25), and  $q_k = e^{-\lambda_k}$  gives (26). Absolute convergence was already established. For fixed  $K$ , the omitted magnitudes form a convergent sequence dominated by its first scale times a constant independent of sufficiently large  $n$ :

$$\sum_{k>K} q_k^{n+1} = q_{K+1}^{n+1} \left( 1 + \sum_{k>K+1} \left( \frac{q_k}{q_{K+1}} \right)^{n+1} \right) = \mathcal{O}_K(q_{K+1}^{n+1}).$$

This proves (27). □

The first few sector parameters are displayed in Table 1.

Table 1: The first exponential–oscillatory sectors. Here  $q_k = e^{-\lambda_k}$  and  $\theta_k/\pi$  is the normalized phase.

| $k$ | $r_k$         | $q_k$        | $\lambda_k$  | $\theta_k/\pi$ |
|-----|---------------|--------------|--------------|----------------|
| 1   | 6.3213029209  | 0.1096525810 | 2.2104382659 | 0.4650261734   |
| 2   | 12.5854727139 | 0.0550751804 | 2.8990561102 | 0.4824601508   |
| 3   | 18.8622960281 | 0.0367477628 | 3.3036779310 | 0.4883001896   |
| 4   | 25.1422977208 | 0.0275689672 | 3.5910645166 | 0.4912234132   |
| 5   | 31.4235722527 | 0.0220581917 | 3.8140712405 | 0.4929780900   |
| 6   | 37.7054835106 | 0.0183831930 | 3.9963184557 | 0.4941481183   |

In particular, the first exponentially improved approximation is

$$F_n = \frac{n!}{2\alpha^{n+1}} \{1 + 2q_1^{n+1} \cos((n+1)\theta_1) + \mathcal{O}(q_2^{n+1})\}. \quad (28)$$

The correction oscillates because the nearest nonreal singularities occur as a conjugate pair.

## 4.2 Why every algebraic correction vanishes after factorial normalization

**Corollary 4.2** (Flatness beyond all powers). *For every fixed  $M \geq 1$ ,*

$$\frac{2\alpha^{n+1}}{n!} F_n = 1 + o(n^{-M}). \quad (29)$$

*Equivalently, the complete Poincaré expansion of the normalized ratio is*

$$\frac{2\alpha^{n+1}}{n!} F_n \sim 1 + \frac{0}{n} + \frac{0}{n^2} + \frac{0}{n^3} + \cdots. \quad (30)$$

*Proof.* By (27) with  $K = 0$ , the difference from 1 is  $\mathcal{O}(q_1^n)$ , where  $0 < q_1 < 1$ . Exponential decay dominates every fixed inverse power of  $n$ .  $\square$

#### Structural point

A pole of order  $s$  contributes a polynomial of degree  $s - 1$  in  $n$  times a geometric term. Here every pole is simple, so  $s = 1$ : the polynomial has degree zero. This is the analytic reason that the factorial-normalized Fubini transseries has one coefficient per pole and no  $1/n$  descendants. The nonzero inverse-power coefficients that appear later come solely from choosing to expand the exact factor  $n!$ .

### 4.3 Coefficient dictionary in the natural transseries scale

If the transmonomial of sector  $k$  is taken to be  $\mathbf{m}_k(n) := e^{-(n+1)\Lambda_k} = \omega_k^{n+1}$ , then (24) reads

$$\frac{2\alpha^{n+1}}{n!} F_n = \sum_{k \in \mathbb{Z}} \mathbf{m}_k(n). \quad (31)$$

Thus the full coefficient array is particularly simple:

$$c_{k,0} = 1 \quad (k \in \mathbb{Z}), \quad c_{k,j} = 0 \quad (k \in \mathbb{Z}, j \geq 1). \quad (32)$$

This should not be mistaken for a lack of structure. All geometry has moved into the transmonomials themselves, through their exact complex actions  $\Lambda_k$ .

## 5 Certified truncation and remainder bounds

### 5.1 The exact remainder after finitely many pole pairs

Define the  $K$ -pair approximation

$$S_K(n) := \frac{n!}{2} \sum_{k=-K}^K \rho_k^{-n-1} = \frac{n!}{2\alpha^{n+1}} \left[ 1 + 2 \sum_{k=1}^K q_k^{n+1} \cos((n+1)\theta_k) \right]. \quad (33)$$

**Theorem 5.1** (Hurwitz-zeta tail bound). *For  $n \geq 1$  and  $K \geq 0$ ,*

$$|F_n - S_K(n)| \leq n! \sum_{k=K+1}^{\infty} r_k^{-n-1} \quad (34)$$

$$\leq \frac{n!}{(2\pi)^{n+1}} \zeta(n+1, K+1), \quad (35)$$

where  $\zeta(s, a) = \sum_{j=0}^{\infty} (j+a)^{-s}$  is the Hurwitz zeta function. In elementary terms,

$$|F_n - S_K(n)| \leq \frac{n!}{(2\pi)^{n+1}} \left( (K+1)^{-n-1} + \frac{(K+1)^{-n}}{n} \right). \quad (36)$$

Relative to the dominant term  $A_n = n!/(2\alpha^{n+1})$ ,

$$\frac{|F_n - S_K(n)|}{A_n} \leq 2 \left( \frac{\alpha}{2\pi} \right)^{n+1} \zeta(n+1, K+1). \quad (37)$$

*Proof.* Pairing the omitted indices  $k$  and  $-k$  in (15) and applying the triangle inequality gives (34). Since  $r_k = \sqrt{\alpha^2 + 4\pi^2 k^2} \geq 2\pi k$ , equation (35) follows. The integral test gives

$$\sum_{k=K+1}^{\infty} k^{-n-1} \leq (K+1)^{-n-1} + \int_{K+1}^{\infty} x^{-n-1} dx,$$

which is (36). Dividing (35) by  $A_n$  gives (37).  $\square$

Taking  $K = 0$  gives the explicit leading-term estimate

$$\left| \frac{F_n}{A_n} - 1 \right| \leq 2 \left( \frac{\alpha}{2\pi} \right)^{n+1} \zeta(n+1). \quad (38)$$

This proves the classical equivalent (3) with a fully explicit, exponentially small relative error.

## 5.2 Relative accuracy versus absolute accuracy

Because  $F_n$  grows factorially, an approximation can have a spectacularly small relative error and still miss the exact integer by a huge amount. The numerical values in Table 2 illustrate this distinction. Here  $S_0$  uses only the real pole  $\rho_0$ , while  $S_1$  also uses the first conjugate pair.

Table 2: Absolute errors of pole truncations and relative error of the leading term. Values are rounded for display.

| $n$ | $ S_0(n) - F_n $      | $ S_1(n) - F_n $      | $ S_0(n)/F_n - 1 $     |
|-----|-----------------------|-----------------------|------------------------|
| 5   | $1.52 \times 10^{-3}$ | $3.18 \times 10^{-5}$ | $2.81 \times 10^{-6}$  |
| 10  | $5.27 \times 10^{-3}$ | $1.66 \times 10^{-6}$ | $5.16 \times 10^{-11}$ |
| 16  | $4.87 \times 10^{-1}$ | $3.38 \times 10^{-6}$ | $9.16 \times 10^{-17}$ |
| 17  | $5.42 \times 10^{-1}$ | $3.10 \times 10^{-6}$ | $4.16 \times 10^{-18}$ |
| 20  | $2.75 \times 10^1$    | $1.78 \times 10^{-5}$ | $1.03 \times 10^{-20}$ |
| 30  | $1.04 \times 10^7$    | $2.11 \times 10^{-2}$ | $9.10 \times 10^{-31}$ |
| 50  | $2.76 \times 10^{23}$ | $7.98 \times 10^7$    | $1.38 \times 10^{-49}$ |

The leading term rounds correctly through  $n = 16$ , but at  $n = 17$  its absolute error already exceeds  $1/2$ . This does not contradict asymptotic equivalence: the relative error is then approximately  $4.16 \times 10^{-18}$ .

## 5.3 Certified nearest-integer recovery

**Corollary 5.2** (A rigorous rounding certificate). *If*

$$\frac{n!}{(2\pi)^{n+1}} \left( (K+1)^{-n-1} + \frac{(K+1)^{-n}}{n} \right) < \frac{1}{2}, \quad (39)$$

*then  $F_n$  is the unique nearest integer to  $S_K(n)$ .*

*Proof.* The left side of (39) is an upper bound for  $|F_n - S_K(n)|$  by (36). Since  $F_n$  is an integer, distance less than  $1/2$  identifies it uniquely by rounding.  $\square$

**Proposition 5.3** (A sufficient linear pole budget). *Fix  $\delta > 0$ . If  $K = K_n$  is chosen so that*

$$K_n + 1 \geq \frac{(1+\delta)n}{2\pi e}, \quad (40)$$

*then the certified bound in (36) is*

$$\mathcal{O}_\delta \left( n^{-1/2} (1+\delta)^{-n} \right). \quad (41)$$

*In particular, (39) holds for all sufficiently large  $n$ .*

*Proof.* Use the elementary upper form of Stirling's estimate  $n! \leq C\sqrt{n}(n/e)^n$  with an absolute constant  $C$ . Put  $L = K_n + 1$ . The two terms in (36) are bounded by constant multiples of

$$\frac{\sqrt{n}}{L} \left( \frac{n}{2\pi e L} \right)^n \quad \text{and} \quad \frac{1}{\sqrt{n}} \left( \frac{n}{2\pi e L} \right)^n,$$

respectively. Condition (40) makes the parenthesized ratio at most  $(1 + \delta)^{-1}$ . It also gives  $L \geq (1 + \delta)n/(2\pi e)$ , hence  $\sqrt{n}/L = \mathcal{O}_\delta(n^{-1/2})$ . This proves (41).  $\square$

Thus only about  $n/(2\pi e) \approx 0.05855n$  conjugate pole pairs are needed for a simple asymptotic rounding certificate. This is a sufficient asymptotic budget, not a claim of pointwise minimality: oscillatory cancellation often allows fewer pairs.

## 6 The universal Stirling descendant tower

### 6.1 From the exact factorial prefactor to an elementary scale

The natural formula (15) keeps  $n!$  exact. To rewrite it in the more conventional elementary scale, define

$$P_n := \frac{\sqrt{2\pi n}}{2\alpha} \left( \frac{n}{e\alpha} \right)^n. \quad (42)$$

Stirling's expansion [5, §5.11] has the form

$$n! \sim \sqrt{2\pi n} \left( \frac{n}{e} \right)^n \sum_{j=0}^{\infty} \frac{\gamma_j}{n^j}. \quad (43)$$

The coefficients  $\gamma_j$  are universal: they know nothing about Fubini numbers. We now derive them in a form suited to coefficient calculus.

### 6.2 Bernoulli exponent and Bell-polynomial formula

Use the Bernoulli convention

$$\frac{t}{e^t - 1} = \sum_{m=0}^{\infty} B_m \frac{t^m}{m!}, \quad B_1 = -\frac{1}{2}. \quad (44)$$

The logarithmic Stirling expansion is

$$\log n! \sim \left( n + \frac{1}{2} \right) \log n - n + \frac{1}{2} \log(2\pi) + \sum_{r=1}^{\infty} \frac{a_r}{n^r}, \quad (45)$$

where

$$a_r := \frac{B_{r+1}}{r(r+1)}. \quad (46)$$

Thus  $a_1 = 1/12$ ,  $a_2 = 0$ ,  $a_3 = -1/360$ , and all even  $a_r$  with  $r \geq 2$  vanish.

Let  $Y_j(x_1, \dots, x_j)$  denote the complete exponential Bell polynomial, defined by

$$\exp \left( \sum_{r \geq 1} x_r \frac{t^r}{r!} \right) = \sum_{j \geq 0} Y_j(x_1, \dots, x_j) \frac{t^j}{j!}. \quad (47)$$

**Theorem 6.1** (General Stirling coefficient formulas). *The coefficients in (43) are determined by*

$$\sum_{j \geq 0} \gamma_j t^j = \exp \left( \sum_{r \geq 1} a_r t^r \right). \quad (48)$$

Equivalently, for every  $j \geq 0$ ,

$$\gamma_j = \frac{1}{j!} Y_j(1!a_1, 2!a_2, \dots, j!a_j), \quad (49)$$

$$= \sum_{\substack{m_1, \dots, m_j \geq 0 \\ m_1 + 2m_2 + \dots + jm_j = j}} \prod_{r=1}^j \frac{a_r^{m_r}}{m_r!}, \quad (50)$$

$$\gamma_0 = 1, \quad \gamma_j = \frac{1}{j} \sum_{r=1}^j \frac{B_{r+1}}{r+1} \gamma_{j-r} \quad (j \geq 1). \quad (51)$$

*Proof.* Exponentiating the correction series in (45) gives (48). Substitution  $x_r = r!a_r$  in (47) gives (49). Expanding the exponential as a product  $\prod_{r \geq 1} \exp(a_r t^r)$  and extracting  $t^j$  gives (50). Finally, logarithmic differentiation of (48) gives

$$\frac{\sum_{j \geq 1} j \gamma_j t^{j-1}}{\sum_{j \geq 0} \gamma_j t^j} = \sum_{r \geq 1} r a_r t^{r-1}.$$

Comparing coefficients of  $t^{j-1}$  and using  $ra_r = B_{r+1}/(r+1)$  proves (51).  $\square$

The first coefficients are listed in Table 3.

Table 3: Stirling coefficients  $\gamma_j$  in (43).

| $j$ | $\gamma_j$                 | $j$ | $\gamma_j$                                      |
|-----|----------------------------|-----|-------------------------------------------------|
| 0   | 1                          | 6   | $\frac{5246819}{75246796800}$                   |
| 1   | $\frac{1}{12}$             | 7   | $-\frac{534703531}{902961561600}$               |
| 2   | $\frac{1}{288}$            | 8   | $-\frac{4483131259}{86684309913600}$            |
| 3   | $-\frac{139}{51840}$       | 9   | $\frac{432261921612371}{514904800886784000}$    |
| 4   | $-\frac{571}{2488320}$     | 10  | $\frac{6232523202521089}{86504006548979712000}$ |
| 5   | $\frac{163879}{209018880}$ |     |                                                 |

### 6.3 The full two-level coefficient theorem

**Theorem 6.2** (All pole-sector and algebraic coefficients). *Let  $P_n$  be given by (42), and let  $\Lambda_k = \text{Log}(\rho_k/\alpha)$ . Then the Fubini numbers have the formal two-level transseries*

$$F_n \sim P_n \sum_{k \in \mathbb{Z}} e^{-n\Lambda_k} \sum_{j=0}^{\infty} \frac{C_{k,j}}{n^j} \quad (52)$$

with the explicit general coefficient

$$C_{k,j} = e^{-\Lambda_k} \gamma_j = \frac{\alpha}{\alpha + 2\pi i k} \frac{1}{j!} Y_j(1!a_1, \dots, j!a_j). \quad (53)$$

Equivalently,

$$C_{k,j} = \frac{\alpha}{\alpha + 2\pi i k} \sum_{\substack{m_1, \dots, m_j \geq 0 \\ \sum_{r=1}^j r m_r = j}} \prod_{r=1}^j \frac{1}{m_r!} \left( \frac{B_{r+1}}{r(r+1)} \right)^{m_r}. \quad (54)$$

*Proof.* Write (15) as

$$F_n = \frac{n!}{2\alpha^{n+1}} \sum_{k \in \mathbb{Z}} e^{-(n+1)\Lambda_k}.$$

Insert (43), and observe that

$$\frac{\sqrt{2\pi n}(n/e)^n}{2\alpha^{n+1}} = P_n, \quad e^{-(n+1)\Lambda_k} = e^{-n\Lambda_k} e^{-\Lambda_k}.$$

The coefficient of  $e^{-n\Lambda_k} n^{-j}$  is therefore  $e^{-\Lambda_k} \gamma_j$ , which is (53). Substituting (50) gives (54).  $\square$

Pairing conjugates gives the real form

$$F_n \sim P_n \left( \sum_{j \geq 0} \frac{\gamma_j}{n^j} \right) \left[ 1 + 2 \sum_{k \geq 1} q_k^{n+1} \cos((n+1)\theta_k) \right]. \quad (55)$$

The bracketed factor is exact; only the Stirling factor is asymptotic.

### 6.4 A precise double-truncation statement

For integers  $J \geq 1$  and  $K \geq 0$ , define

$$T_{J,K}(n) := P_n \left( \sum_{j=0}^{J-1} \frac{\gamma_j}{n^j} \right) \sum_{k=-K}^K e^{-(n+1)\Lambda_k}. \quad (56)$$

For fixed  $J, K$ , the standard remainder statement for Stirling's expansion and Theorem 5.1 give

$$F_n - T_{J,K}(n) = \mathcal{O}_{J,K}(P_n n^{-J}) + \mathcal{O}\left(\frac{n!}{(2\pi)^{n+1}} \zeta(n+1, K+1)\right). \quad (57)$$

The first term is the algebraic truncation error of the exact factorial; the second is the omitted Fubini pole tail. The two truncations are independent and should be chosen according to the desired task. For relative asymptotics, a small fixed  $K$  is usually enough. For exact integer reconstruction,  $K$  must grow with  $n$  as discussed in Theorem 5.3.

## 7 A multi-index coefficient calculus for nonlinear observables

The linear sequence  $F_n$  itself contains one copy of each pole sector. Products, powers, logarithms, reciprocals, and ratios generate sums of pole actions. This section gives their coefficients in closed form.

### 7.1 The normalized exponential correction

Put

$$H_n := \frac{2\alpha^{n+1}}{n!} F_n = 1 + E_n, \quad E_n := \sum_{k \neq 0} e^{-(n+1)\Lambda_k}. \quad (58)$$

By Theorem 4.2,  $E_n \rightarrow 0$  exponentially. Let  $\mathcal{I}$  be the set of finitely supported multi-indices  $\mathbf{m} = (m_k)_{k \in \mathbb{Z} \setminus \{0\}}$  with  $m_k \in \mathbb{N}$ . Define

$$|\mathbf{m}| := \sum_{k \neq 0} m_k, \quad \Lambda_{\mathbf{m}} := \sum_{k \neq 0} m_k \Lambda_k, \quad \mathbf{m}! := \prod_{k \neq 0} m_k!. \quad (59)$$

All sums and products are finite for each  $\mathbf{m}$ .

**Theorem 7.1** (Analytic-function coefficient rule). *Let  $\Phi$  be analytic in a neighborhood of 1. Formally, and analytically for all sufficiently large  $n$ ,*

$$\Phi(H_n) = \sum_{\mathbf{m} \in \mathcal{I}} \frac{\Phi^{(|\mathbf{m}|)}(1)}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}. \quad (60)$$

The zero multi-index contributes  $\Phi(1)$ .

*Proof.* Taylor-expand  $\Phi(1 + E_n)$ :

$$\Phi(1 + E_n) = \sum_{\ell \geq 0} \frac{\Phi^{(\ell)}(1)}{\ell!} E_n^\ell.$$

The multinomial theorem gives

$$E_n^\ell = \sum_{\substack{\mathbf{m} \in \mathcal{I} \\ |\mathbf{m}| = \ell}} \frac{\ell!}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}.$$

Substitution proves (60). Since  $E_n \rightarrow 0$ , the Taylor series converges analytically for sufficiently large  $n$ .  $\square$

This theorem is the transseries analogue of the ordinary multinomial coefficient calculus. It also displays a useful distinction: the primitive Fubini sequence has only single-pole sectors, while nonlinear observables have all additive combinations  $\Lambda_{\mathbf{m}}$ .

### 7.2 Arbitrary complex powers

For  $\Phi(u) = u^\beta$ , where a branch is fixed near  $u = 1$ ,  $\Phi^{(\ell)}(1) = \beta^\ell$ . Hence

$$H_n^\beta = \sum_{\mathbf{m} \in \mathcal{I}} \frac{\beta^{|\mathbf{m}|}}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}. \quad (61)$$

For  $\beta = 1$ , the falling factorial vanishes for  $|\mathbf{m}| \geq 2$ ; this recovers the fact that  $H_n$  contains only the vacuum and one-pole sectors.

The algebraic Stirling factor also has an all-orders power calculus. Define

$$\sum_{j \geq 0} \gamma_j^{(\beta)} t^j := \exp \left( \beta \sum_{r \geq 1} a_r t^r \right). \quad (62)$$

Then

$$\gamma_j^{(\beta)} = \frac{1}{j!} Y_j(\beta 1! a_1, \dots, \beta j! a_j), \quad (63)$$

$$= \sum_{\substack{m_1, \dots, m_j \geq 0 \\ \sum r m_r = j}} \beta^{m_1 + \dots + m_j} \prod_{r=1}^j \frac{a_r^{m_r}}{m_r!}. \quad (64)$$

Combining (61) and (62) yields the complete power transseries.

**Corollary 7.2** (All coefficients of  $F_n^\beta$ ). *With  $P_n$  as in (42),*

$$F_n^\beta \sim P_n^\beta \left( \sum_{j \geq 0} \frac{\gamma_j^{(\beta)}}{n^j} \right) \sum_{\mathbf{m} \in \mathcal{I}} \frac{\beta^{|\mathbf{m}|}}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}. \quad (65)$$

Consequently, the coefficient at algebraic order  $j$  and composite action  $\Lambda_{\mathbf{m}}$  is

$$C_{\mathbf{m},j}^{(\beta)} = \gamma_j^{(\beta)} \frac{\beta^{|\mathbf{m}|}}{\mathbf{m}!}. \quad (66)$$

If the transmonomial is written as  $e^{-n\Lambda_{\mathbf{m}}}$  rather than  $e^{-(n+1)\Lambda_{\mathbf{m}}}$ , the coefficient is multiplied by  $e^{-\Lambda_{\mathbf{m}}}$ .

### 7.3 The logarithm

For  $\Phi(u) = \log u$ ,  $\Phi^{(\ell)}(1) = (-1)^{\ell-1}(\ell-1)!$  for  $\ell \geq 1$ . Thus

$$\log H_n = \sum_{\substack{\mathbf{m} \in \mathcal{I} \\ \mathbf{m} \neq \mathbf{0}}} (-1)^{|\mathbf{m}|+1} \frac{(|\mathbf{m}|-1)!}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}. \quad (67)$$

Since the logarithm of the Stirling factor is exactly the Bernoulli exponent in (45), we obtain

$$\begin{aligned} \log F_n &\sim \log P_n + \sum_{r \geq 1} \frac{a_r}{n^r} \\ &\quad + \sum_{\substack{\mathbf{m} \in \mathcal{I} \\ \mathbf{m} \neq \mathbf{0}}} (-1)^{|\mathbf{m}|+1} \frac{(|\mathbf{m}|-1)!}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}. \end{aligned} \quad (68)$$

Equations (46) and (68) give every algebraic and every composite exponential coefficient of  $\log F_n$ .

### 7.4 Reciprocals and ratios

For the reciprocal,  $\Phi(u) = u^{-1}$ , equation (60) gives

$$\frac{1}{H_n} = \sum_{\mathbf{m} \in \mathcal{I}} (-1)^{|\mathbf{m}|} \frac{|\mathbf{m}|!}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}. \quad (69)$$



An exact ratio identity follows directly from (58):

$$\frac{nF_{n-1}}{F_n} = \alpha \frac{H_{n-1}}{H_n}. \quad (70)$$

Since  $H_{n-1} = 1 + \sum_{k \neq 0} \omega_k^n$ , linearization gives

$$\frac{nF_{n-1}}{\alpha F_n} = 1 + \sum_{k \neq 0} (1 - \omega_k) \omega_k^n + \mathcal{O}(q_1^{2n}). \quad (71)$$

Keeping only the nearest conjugate pair gives

$$\frac{nF_{n-1}}{\alpha F_n} = 1 + 2q_1^n \Re \left( (1 - \omega_1) e^{-in\theta_1} \right) + \mathcal{O}(q_2^n). \quad (72)$$

In particular,  $nF_{n-1}/F_n \rightarrow \log 2$ , with an explicitly oscillatory exponential approach.

## 8 Borel transform and resurgent large-order structure

### 8.1 The divergent ordinary series and its exact Borel transform

The ordinary generating series

$$\widehat{\mathcal{F}}(x) := \sum_{n=0}^{\infty} F_n x^{n+1} \quad (73)$$

has zero radius of convergence because  $F_n$  grows factorially. It is, however, Gevrey of order one. With the convention

$$\mathcal{B} \left( \sum_{n \geq 0} a_n x^{n+1} \right) (\xi) := \sum_{n \geq 0} a_n \frac{\xi^n}{n!}, \quad (74)$$

we obtain immediately from (4):

**Theorem 8.1** (Exact Borel transform).

$$\boxed{\mathcal{B}\widehat{\mathcal{F}}(\xi) = \frac{1}{2 - e^\xi}}. \quad (75)$$

*It is meromorphic on  $\mathbb{C}$ , with simple singularities  $\xi = \rho_k$  and residue  $-1/2$  at every one.*

The singularity dictionary is therefore exact:

| Borel datum     | value                                   | large-order contribution |
|-----------------|-----------------------------------------|--------------------------|
| singularity     | $\rho_k = \alpha + 2\pi i k$            | sector $k$               |
| residue         | $-1/2$                                  | $+n!/(2\rho_k^{n+1})$    |
| relative action | $\Lambda_k = \text{Log}(\rho_k/\alpha)$ | $e^{-(n+1)\Lambda_k}$    |

Summing the large-order contribution over all Borel singularities reproduces (15) exactly. In this example, the usual resurgent “large-order relation” closes without an uncomputed remainder.

## 8.2 Lateral Borel–Laplace sums and Stokes jumps

For a direction  $\vartheta$  not containing a pole, define formally

$$\mathcal{S}_\vartheta \widehat{\mathcal{F}}(x) := \int_0^{e^{i\vartheta}\infty} e^{-\xi/x} \frac{d\xi}{2 - e^\xi}, \quad (76)$$

whenever the integral converges. Expanding the Borel transform at the origin and integrating term by term recovers (73) asymptotically. The positive real direction is a Stokes direction because it contains  $\rho_0 = \alpha$ . With  $\mathcal{S}_{0+}$  and  $\mathcal{S}_{0-}$  denoting paths just above and below the positive axis, our orientation convention gives

$$\mathcal{S}_{0+} \widehat{\mathcal{F}}(x) - \mathcal{S}_{0-} \widehat{\mathcal{F}}(x) = \pi i e^{-\alpha/x}. \quad (77)$$

Indeed, the jump is  $-2\pi i$  times the residue of  $e^{-\xi/x} (2 - e^\xi)^{-1}$  at  $\xi = \alpha$ , and that residue is  $-\frac{1}{2} e^{-\alpha/x}$ . Analogous jumps occur on rays through the other  $\rho_k$ . Standard background on Borel summation and resurgence can be found in Costin [4] and Sauzin [11].

*Remark 8.2* (Two different transseries variables). The large-order transseries studied in the preceding sections uses  $n \rightarrow \infty$  and actions  $\Lambda_k$ . The Borel–Laplace problem uses  $x \rightarrow 0$  and actions  $\rho_k/x$ . They encode the same pole lattice in two dual ways:  $\rho_k^{-n-1}$  controls Taylor-coefficient growth, while  $e^{-\rho_k/x}$  controls Stokes phenomena of the Borel sum.

## 9 Fubini polynomials and a one-parameter pole lattice

### 9.1 Definition and geometric-moment representation

Define the Fubini polynomials

$$F_n(y) := \sum_{j=0}^n j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} y^j. \quad (78)$$

Their exponential generating function is

$$\sum_{n \geq 0} F_n(y) \frac{z^n}{n!} = \frac{1}{1 - y(e^z - 1)} = \frac{1}{1 + y - ye^z}. \quad (79)$$

For  $y > 0$ , set

$$\alpha_y := \log \left( 1 + \frac{1}{y} \right), \quad \rho_k(y) := \alpha_y + 2\pi i k. \quad (80)$$

Expanding (79) geometrically gives

$$F_n(y) = \frac{1}{1+y} \sum_{m=0}^{\infty} \left( \frac{y}{1+y} \right)^m m^n. \quad (81)$$

**Theorem 9.1** (Complete transseries of the Fubini polynomials). *For  $y > 0$  and  $n \geq 1$ ,*

$$F_n(y) = \frac{n!}{1+y} \sum_{k \in \mathbb{Z}} \rho_k(y)^{-n-1}. \quad (82)$$

Consequently,

$$F_n(y) = \frac{n!}{(1+y)\alpha_y^{n+1}} \left[ 1 + 2 \sum_{k \geq 1} \left( \frac{\alpha_y}{\sqrt{\alpha_y^2 + 4\pi^2 k^2}} \right)^{n+1} \times \cos \left( (n+1) \arctan \frac{2\pi k}{\alpha_y} \right) \right]. \quad (83)$$

All coefficient and remainder formulas of the preceding sections remain valid after replacing  $\alpha$  by  $\alpha_y$  and  $1/2$  by  $1/(1+y)$ .

*Proof.* Equation (81) is the moment sum with decay parameter  $\alpha_y$ , because  $y/(1+y) = e^{-\alpha_y}$ . Applying the proof of Theorem 3.2 gives (82); pairing conjugates gives (83).  $\square$

The ordinary Fubini numbers are recovered at  $y = 1$ , for which  $\alpha_y = \log 2$  and  $1/(1+y) = 1/2$ .

## 10 Higher-order poles and Nörlund coefficient calculus

The simplicity of the Fubini transseries is caused by simple poles. It is useful to see the general pattern when those poles are raised to order  $r$ . This also supplies a nontrivial family of all-orders pole coefficients.

### 10.1 Convolution powers

For an integer  $r \geq 1$ , define

$$A_n^{(r)} := n! [z^n] (2 - e^z)^{-r}. \quad (84)$$

Combinatorially,  $A_n^{(r)}$  is the  $r$ -fold binomial convolution of the Fubini sequence, because products of exponential generating functions encode binomial convolution.

Define the generalized Bernoulli, or Nörlund, numbers by

$$\left( \frac{t}{e^t - 1} \right)^r = \sum_{j=0}^{\infty} B_j^{(r)} \frac{t^j}{j!}. \quad (85)$$

For example,  $B_0^{(r)} = 1$ ,  $B_1^{(r)} = -r/2$ , and  $B_2^{(r)} = r(3r-1)/12$ .

**Theorem 10.1** (General coefficient formula for order- $r$  pole sectors). *For  $r \geq 1$  and  $n \geq 1$ ,*

$$A_n^{(r)} = \frac{n!}{2^r} \sum_{k \in \mathbb{Z}} \sum_{j=0}^{r-1} \frac{(-1)^j B_j^{(r)}}{j!} \binom{n+r-j-1}{r-j-1} \rho_k^{-n-r+j}. \quad (86)$$

Thus each pole sector is a polynomial in  $n$  of degree  $r-1$ , multiplied by the geometric factor  $\rho_k^{-n}$ . Formula (15) is the case  $r = 1$ .

*Proof.* We first justify that the coefficient is determined exactly by the principal parts at the full pole lattice. Apply the residue theorem to

$$\frac{(2 - e^z)^{-r}}{z^{n+1}}$$

on rectangles whose vertical sides are  $\Re z = \pm R$  and whose horizontal sides are  $\Im z = \pm 2\pi(N + \frac{1}{2})$ , with  $R, N \rightarrow \infty$  in a comparable way. On the left side the denominator  $2 - e^z$  stays bounded away from zero; on the right side its reciprocal decays exponentially; and on the horizontal sides  $e^z = -e^{\Re z}$ , so again the denominator stays away from zero. Since  $n \geq 1$ , the four boundary integrals therefore tend to zero. Consequently, the residue at the origin, which is  $[z^n] (2 - e^z)^{-r}$ , equals the negative of the sum of the residues at all  $\rho_k$ .

Near a pole  $\rho_k$ , put  $t = z - \rho_k$ . Since  $e^{\rho_k} = 2$ ,

$$\begin{aligned} (2 - e^z)^{-r} &= (-2)^{-r} (e^t - 1)^{-r} \\ &= (-2)^{-r} t^{-r} \left( \frac{t}{e^t - 1} \right)^r \\ &= (-2)^{-r} \sum_{j \geq 0} \frac{B_j^{(r)}}{j!} t^{j-r}. \end{aligned} \quad (87)$$

Only  $j = 0, 1, \dots, r - 1$  belong to the principal part. If  $m = r - j$ , then

$$[z^n](z - \rho_k)^{-m} = (-1)^m \binom{n + m - 1}{m - 1} \rho_k^{-n-m}. \quad (88)$$

This coefficient is precisely the negative of the residue contribution of  $(z - \rho_k)^{-m} z^{-n-1}$ , in agreement with the preceding contour identity. Multiplying (88) by the coefficient of  $t^{-m}$  in (87) gives the inner summand of (86); the signs simplify to  $(-1)^j$ . Finally, every term is  $\mathcal{O}(|k|^{-n-1})$  or smaller, so the pole sum is absolutely convergent for  $n \geq 1$ , completing the proof.  $\square$

For  $r = 2$ , one has  $B_0^{(2)} = 1$  and  $B_1^{(2)} = -1$ , so

$$A_n^{(2)} = \frac{n!}{4} \sum_{k \in \mathbb{Z}} [(n+1)\rho_k^{-n-2} + \rho_k^{-n-1}]. \quad (89)$$

The degree-one polynomial in  $n$  is exactly what a double pole predicts. This comparison isolates the special feature of the original Fubini sequence: its degree-zero sector polynomial is not an approximation but the whole local contribution.

## 11 Numerical verification and reproducible computation

### 11.1 Exact arithmetic checks

For  $n \geq 1$ , the three exact representations

$$F_n = \sum_{j=0}^n j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} = \frac{1}{2} \sum_{m \geq 0} \frac{m^n}{2^m} = \frac{n!}{2} \sum_{k \in \mathbb{Z}} \rho_k^{-n-1} \quad (90)$$

serve different computational purposes.

- The finite Stirling sum is ideal for exact integer computation.
- The geometric moment sum is positive and conceptually simple, but its efficient truncation depends on locating the maximum near  $m = n/\alpha$ .
- The pole sum converges extraordinarily fast in relative terms as  $n$  grows and comes with the certified bound (35).

The following Wolfram Language fragment implements the finite definition, the pole approximation, and the rigorous elementary tail bound.

```
ClearAll[fubini, rho, poleApproximation, poleTailBound];

fubini[n_Integer?NonNegative] :=
  Sum[k! StirlingS2[n, k], {k, 0, n}];

rho[k_Integer] := Log[2] + 2 Pi I k;

poleApproximation[n_Integer?Positive, K_Integer?NonNegative] :=
  n!/2 Sum[rho[k]^(-n - 1), {k, -K, K}];

poleTailBound[n_Integer?Positive, K_Integer?NonNegative] :=
  n!/(2 Pi)^(n + 1) ((K + 1)^(-n - 1) +
    (K + 1)^(-n)/n);
```

A certified exact-recovery routine increases  $K$  until  $\text{poleTailBound}[n, K] < 1/2$ , evaluates the symmetric pole sum with enough guard digits, and rounds its real part. The imaginary part should vanish up to numerical error because the summands occur in conjugate pairs.

## 11.2 Stable real evaluation

For numerical work it is preferable to use the explicitly real formula (33) rather than summing complex powers independently. A stable implementation is

```
ClearAll[realPoleApproximation];

realPoleApproximation[n_Integer?Positive,
                      K_Integer?NonNegative] := Module[
  {a = Log[2]},
  n!/(2 a^(n + 1)) (1 +
    2 Sum[
      (a/Sqrt[a^2 + 4 Pi^2 k^2])^(n + 1) *
      Cos[(n + 1) ArcTan[2 Pi k/a]],
      {k, 1, K}])
  ];
```

The scale separation in  $q_k^{n+1}$  means that high working precision may be necessary when the goal is exact rounding rather than relative approximation. One should request enough digits to resolve the absolute tail target, not just the relative size of  $F_n$ .

## 11.3 Generation of Stirling descendants

The recurrence (51) gives a direct exact algorithm:

```
ClearAll[stirlingGamma];
stirlingGamma[0] = 1;
stirlingGamma[j_Integer?Positive] := stirlingGamma[j] =
  1/j Sum[BernoulliB[r + 1]/(r + 1) *
    stirlingGamma[j - r], {r, 1, j}];
```

Alternatively, the complete Bell-polynomial formula (49) can be used directly. Both calculations are exact over the rationals.

# 12 Consequences, comparisons, and interpretation

## 12.1 A complete hierarchy of approximants

The exact real transseries gives the nested sequence

$$\begin{aligned} F_n &\sim \frac{n!}{2\alpha^{n+1}}, \\ F_n &\sim \frac{n!}{2\alpha^{n+1}} \left[ 1 + 2q_1^{n+1} \cos((n+1)\theta_1) \right], \\ F_n &\sim \frac{n!}{2\alpha^{n+1}} \left[ 1 + 2 \sum_{k=1}^2 q_k^{n+1} \cos((n+1)\theta_k) \right], \end{aligned}$$

and so on. Unlike a divergent power series, this hierarchy is obtained by truncating an absolutely convergent exact sum. There is no least term and no optimal finite truncation for relative accuracy: adding a pole pair always reduces the rigorous absolute tail bound. Optimality enters only through computational cost or through a simultaneous truncation of the divergent Stirling series.

## 12.2 Contrast with ordinary Bell numbers

The ordinary Bell numbers have entire exponential generating function  $\exp(e^z - 1)$ . Their asymptotics arise from moving saddle points and naturally involve the Lambert  $W$ -function and nontrivial

algebraic correction towers. Fubini numbers instead have a meromorphic generating function with a fixed, explicitly known pole lattice. The difference is decisive:

|                                 | Ordinary Bell numbers      | Fubini numbers                    |
|---------------------------------|----------------------------|-----------------------------------|
| <b>EGF</b>                      | $\exp(e^z - 1)$ , entire   | $(2 - e^z)^{-1}$ , meromorphic    |
| <b>Mechanism</b>                | moving saddle point        | fixed singularity lattice         |
| <b>Natural special function</b> | Lambert $W$                | elementary logarithms             |
| <b>Normalized corrections</b>   | nontrivial powers of $1/n$ | pure exponentials, no power tower |
| <b>Full sector data</b>         | saddle expansion           | exact residues of all poles       |

This makes the Fubini sequence a particularly clean model for learning how singularity lattices generate transseries.

### 12.3 Completeness of the result

The word “complete” can be justified concretely.

1. The Borel transform  $(2 - e^\xi)^{-1}$  is known exactly on the whole plane and has no singularities other than  $\rho_k$ .
2. Every singularity is included in (15) with its exact residue contribution.
3. The pole sum is absolutely convergent for every  $n \geq 1$ , so no unspecified summation prescription is needed.
4. The conventional inverse-power descendants introduced by expanding  $n!$  have the explicit coefficient formulas (49)–(51).
5. All additive composite sectors created by nonlinear operations have the multi-index coefficients in (60), (66), and (67).
6. Every finite pole truncation has the certified remainder (35).

There is therefore no hidden Fubini-specific algebraic series, no unlisted exponential action, and no unquantified pole tail.

## 13 Summary of principal formulae

For convenient reference, the main formulas are collected here.

### Notation

$$\alpha = \log 2, \quad \rho_k = \alpha + 2\pi i k, \quad \omega_k = \frac{\alpha}{\rho_k}, \quad \Lambda_k = \text{Log} \frac{\rho_k}{\alpha}.$$

For  $k \geq 1$ ,

$$q_k = \frac{\alpha}{\sqrt{\alpha^2 + 4\pi^2 k^2}}, \quad \lambda_k = -\log q_k, \quad \theta_k = \arctan \frac{2\pi k}{\alpha}.$$

$$F_n = \sum_{j=0}^n j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\}, \quad (91)$$

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (92)$$

$$F_n = \frac{1}{2} \sum_{m \geq 0} \frac{m^n}{2^m}, \quad (93)$$

$$F_n = \frac{n!}{2} \sum_{k \in \mathbb{Z}} \rho_k^{-n-1}, \quad n \geq 1, \quad (94)$$

$$\frac{2\alpha^{n+1}}{n!} F_n = 1 + 2 \sum_{k \geq 1} q_k^{n+1} \cos((n+1)\theta_k), \quad (95)$$

$$|F_n - S_K(n)| \leq \frac{n!}{(2\pi)^{n+1}} \zeta(n+1, K+1), \quad (96)$$

$$F_n \sim \frac{\sqrt{2\pi n}}{2\alpha} \left(\frac{n}{e\alpha}\right)^n \sum_{k \in \mathbb{Z}} e^{-n\Lambda_k} \sum_{j \geq 0} \frac{C_{k,j}}{n^j}, \quad (97)$$

$$C_{k,j} = \frac{\alpha}{\rho_k} \frac{1}{j!} Y_j(1!a_1, \dots, j!a_j), \quad a_r = \frac{B_{r+1}}{r(r+1)}, \quad (98)$$

$$\Phi(H_n) = \sum_{\mathbf{m} \in \mathcal{I}} \frac{\Phi^{(|\mathbf{m}|)}(1)}{\mathbf{m}!} e^{-(n+1)\Lambda_{\mathbf{m}}}, \quad (99)$$

$$\mathcal{BF}(\xi) = \frac{1}{2 - e^\xi}. \quad (100)$$

## A Derivation of the Bell-polynomial coefficient formulas

For completeness, we spell out the coefficient mechanism behind Theorem 6.1. If  $A(t) = \sum_{r \geq 1} a_r t^r$ , then

$$e^{A(t)} = \prod_{r \geq 1} \sum_{m_r \geq 0} \frac{a_r^{m_r} t^{rm_r}}{m_r!}.$$

The coefficient of  $t^j$  is obtained by imposing  $\sum_{r=1}^j rm_r = j$ , which gives (50). This is a partition of the integer  $j$  in frequency form:  $m_r$  records how many parts of size  $r$  appear. The complete exponential Bell polynomial packages the same sum after the rescaling  $x_r = r!a_r$ .

Because  $a_{2s} = 0$  for  $s \geq 1$ , the partition sum may be reduced to odd parts:

$$\gamma_j = \sum_{\substack{m_1, m_3, m_5, \dots \geq 0 \\ \sum_{s \geq 1} (2s-1)m_{2s-1} = j}} \prod_{s \geq 1} \frac{1}{m_{2s-1}!} \left( \frac{B_{2s}}{2s(2s-1)} \right)^{m_{2s-1}}. \quad (101)$$

Only finitely many indices can occur for fixed  $j$ . Formula (101) is often the most efficient closed form for symbolic manipulation.

For example,

$$\begin{aligned} \gamma_3 &= a_3 + \frac{a_1^3}{3!} = -\frac{1}{360} + \frac{1}{6 \cdot 12^3} = -\frac{139}{51840}, \\ \gamma_4 &= a_1 a_3 + \frac{a_1^4}{4!} = -\frac{1}{4320} + \frac{1}{24 \cdot 12^4} = -\frac{571}{2488320}. \end{aligned}$$

These calculations illustrate how Bernoulli data in the logarithm are converted into ordinary Stirling coefficients by set-partition combinatorics.

## B A coefficient-extraction proof of a pole contribution

Suppose a meromorphic generating function has a pole principal part  $A(z - \rho)^{-m}$ , where  $m \geq 1$ . Then

$$\begin{aligned}(z - \rho)^{-m} &= (-\rho)^{-m} \left(1 - \frac{z}{\rho}\right)^{-m} \\ &= (-1)^m \rho^{-m} \sum_{n \geq 0} \binom{n + m - 1}{m - 1} \frac{z^n}{\rho^n}.\end{aligned}$$

Hence

$$[z^n] A(z - \rho)^{-m} = A(-1)^m \binom{n + m - 1}{m - 1} \rho^{-n-m}. \quad (102)$$

For a simple pole,  $m = 1$ , this reduces to  $-A\rho^{-n-1}$ . Since the residue of  $(2 - e^z)^{-1}$  is  $A = -1/2$ , its contribution is  $(2\rho^{n+1})^{-1}$ . Formula (102) also proves the polynomial factor in Theorem 10.1.

## C First values and consistency checks

| $n$ | $F_n$ | $n$ | $F_n$           |
|-----|-------|-----|-----------------|
| 0   | 1     | 8   | 545835          |
| 1   | 1     | 9   | 7087261         |
| 2   | 3     | 10  | 102247563       |
| 3   | 13    | 11  | 1622632573      |
| 4   | 75    | 12  | 28091567595     |
| 5   | 541   | 13  | 526858348381    |
| 6   | 4683  | 14  | 10641342970443  |
| 7   | 47293 | 15  | 230283190977853 |

All displayed values satisfy the following independent checks:

1. the finite Stirling sum (1);
2. the initial condition and recurrence (5);
3. the Taylor coefficients of  $(2 - e^z)^{-1}$ ;
4. the geometric moment sum (6).

For every displayed index  $n \geq 1$ , they also satisfy the pole sum (15), with the numerical truncation error bounded by (35). The redundancy is useful in symbolic and numerical implementations because it separates combinatorial indexing errors from analytic truncation errors.

## Acknowledgment of source context

The combinatorial starting point and notation were aligned with the linked *Combinatorial Coefficient Calculus* in the ProveIt repository, whose ordered-Bell section proves (1), (4), and (5). The organization of algebraic versus exponential scales was informed by the repository's series-and-transseries drafts. The pole transseries, coefficient theorems, tail bounds, nonlinear multi-index calculus, Borel dictionary, and higher-order-pole extension are developed here in a self-contained form.

## References

- [1] R. W. Bailey, The number of weak orderings of a finite set, *Social Choice and Welfare* **15** (1998), 559–562. <https://doi.org/10.1007/s003550050123>.



- [2] J.-P. Barthélemy, An asymptotic equivalent for the number of total preorders on a finite set, *Discrete Mathematics* **29** (1980), 311–313. [https://doi.org/10.1016/0012-365X\(80\)90159-4](https://doi.org/10.1016/0012-365X(80)90159-4).
- [3] L. Comtet, *Advanced Combinatorics: The Art of Finite and Infinite Expansions*, D. Reidel, Dordrecht, 1974.
- [4] O. Costin, *Asymptotics and Borel Summability*, Chapman & Hall/CRC, 2009.
- [5] NIST Digital Library of Mathematical Functions, Chapter 5, especially §5.11, Asymptotic Expansions of the Gamma Function. <https://dlmf.nist.gov/5.11>.
- [6] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [7] OEIS Foundation, Entry A000670, Fubini numbers or ordered Bell numbers. <https://oeis.org/A000670>.
- [8] N. Pippenger, The hypercube of resistors, asymptotic expansions, and preferential arrangements, *Mathematics Magazine* **83** (2010), 331–346. Preprint: <https://arxiv.org/abs/0904.1757>.
- [9] V. Reshetnikov, *Combinatorial Coefficient Calculus: A Unified Formulae Monograph*, ProveIt repository, semi-formalized research-frontier draft, 2026. [https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Combinatorial\\_Coefficient\\_Calculus](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Combinatorial_Coefficient_Calculus).
- [10] V. Reshetnikov, *Series and Transseries*, ProveIt repository, collection of semi-formalized research-frontier drafts, 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>.
- [11] D. Sauzin, Introduction to 1-summability and resurgence, arXiv:1405.0356, 2014. <https://arxiv.org/abs/1405.0356>.
- [12] E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis*, 4th ed., Cambridge University Press, 1927.